

81.6

P.7. Solution: $2xyy' = x^2 + 2y^2$

$$y' = \frac{x}{2y} + \frac{y}{x}$$

$$v = \frac{y}{x}, y = vx, \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1}{2v} + v$$

$$x \frac{dv}{dx} = \frac{1}{2v}$$

$$\int 2v dv = \int \frac{1}{x} dx$$

$$v^2 = \ln|x| + C$$

$$\frac{y^2}{x^2} = \ln|x| + C$$

$$y^2 = x^2 \ln|x| + Cx^2$$

P4. Solution: $(x-y)y' = x+y$

$$y' = \frac{x+y}{x-y}$$

$$\Rightarrow y' = \frac{1 + \frac{y}{x}}{1 - \frac{y}{x}}$$

$$v = \frac{y}{x}, y = vx, \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v}{1-v}$$

$$x \frac{dv}{dx} = \frac{1+v^2}{1-v}$$

$$\int \frac{1-v}{1+v^2} dv = \int \frac{1}{x} dx$$

$$\tan^{-1}v - \frac{1}{2} \ln(1+v^2) = \ln|x| + C$$

$$\tan^{-1}\frac{y}{x} - \frac{1}{2} \ln(1 + \frac{y^2}{x^2}) = \ln|x| + C$$

P7. Solution: $xy^2y' = x^3 + y^3$

$$y' = (\frac{x}{y})^2 + \frac{y}{x}$$

$$v = \frac{y}{x}, y = vx, \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{1}{v^2} + v$$

$$x \frac{dv}{dx} = \frac{1}{v^2}$$

$$\int v^2 dv = \int \frac{1}{x} dx$$

$$\frac{v^3}{3} = \ln|x| + C$$

$$y^3/x^3 = 3(\ln|x| + C)$$

$$y^3 = 3x^3(\ln|x| + C)$$

P12. Solution: $xyy' = y^2 + x\sqrt{4x^2 + y^2}$

$$y' = \frac{y}{x} + \sqrt{4(\frac{y}{x})^2 + 1}$$

$$v = \frac{y}{x}, y = vx, \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = v + \sqrt{\frac{4}{v^2} + 1}$$

$$x \frac{dv}{dx} = \sqrt{\frac{4}{v^2} + 1} = \frac{\sqrt{4+v^2}}{v}$$

$$\int \frac{v}{\sqrt{4+v^2}} dv = \int \frac{1}{x} dx$$

$$\sqrt{4+v^2} = \ln|x| + C$$

$$\sqrt{4 + \frac{y^2}{x^2}} = \ln|x| + C$$

$$x\sqrt{4x^2 + y^2} = \ln|x| + C$$

P16. Solution: $y' = \sqrt{x+y+1}$

$$v = \sqrt{x+y+1}, y = v^2 - x - 1, \frac{dy}{dx} = -1 + 2v \frac{dv}{dx}$$

$$\Rightarrow -1 + 2v \frac{dv}{dx} = v$$

$$2v \frac{dv}{dx} = v + 1$$

$$2 - \frac{2}{v+1} \leftarrow \int \frac{2v}{v+1} dv = \int dx$$

$$\int 2 - \frac{2}{v+1} dv = \int dx$$

$$2v - \ln(v+1)^2 = x + C$$

$$2\sqrt{x+y+1} + \ln(\sqrt{x+y+1})^2 = x + C$$

P17. Solution: $y' = (4x+y)^2$

$$v = 4x+y, y = v-4x, \frac{dy}{dx} = -4 + \frac{dv}{dx}$$

$$\Rightarrow -4 + \frac{dv}{dx} = v^2$$

$$\frac{dv}{dx} = v^2 + 4$$

$$\int \frac{1}{v^2+4} dv = \int dx$$

$$\frac{1}{2} \tan^{-1}(v/2) = x + C$$

$$\frac{v}{2} = \tan(2x+A)$$

$$v = 2 \tan(2x+A) \Rightarrow y = 4x + 2 \tan(2x+A)$$

P22. Solution: $x^2y' + 2xy = 5y^4$

$$y' + \frac{2}{x}y = \frac{5}{x^2}y^4 \rightarrow n=4, 1-n=-3$$

$$v = y^{-3}, y = v^{-\frac{1}{3}}, \frac{dy}{dx} = -\frac{1}{3}v^{-\frac{4}{3}} \frac{dv}{dx}$$

$$-\frac{1}{3}V^{-\frac{2}{3}}\frac{dV}{dx} + \frac{2}{x}V^{-\frac{1}{3}} = \frac{5}{x^2}V^{-\frac{2}{3}}$$

$$\frac{dV}{dx} - \frac{6}{x}V = \frac{15}{x^2}$$

$$P(x) = -\frac{6}{x}, Q(x) = \frac{15}{x^2}$$

$$\Rightarrow g(x) = e^{\int P(x)dx} = e^{-\int \frac{6}{x}dx}$$

$$= e^{-6 \ln|x|} = x^{-6}$$

$$\Rightarrow x^{-6}\frac{dV}{dx} - \frac{6}{x^7}V = \frac{15}{x^6}$$

$$\frac{d}{dx}[x^{-6}V] = \frac{15}{x^6}$$

$$x^{-6}V = \int \frac{15}{x^6}dx = \frac{15}{-5}x^{-5}$$

$$\Rightarrow V(x) = \frac{15}{-5x} + C$$

$$\text{Thus } \left(\frac{1}{y(x)}\right)^3 = \frac{15}{-5x} + C$$

$$\Rightarrow y(x) = \left(\frac{15}{-5x} + C\right)^{-\frac{1}{3}}$$

P26. solution: $3y^2y' + y^3 = e^{-x}$

$$y' + \frac{1}{3}y = \frac{1}{3}e^{-x}y^{-2} \quad (n=-2)$$

$$v = y^{1-n} = y^3, \quad y = v^{\frac{1}{3}}, \quad \frac{dy}{dx} = \frac{1}{3}v^{-\frac{2}{3}}\frac{dv}{dx}$$

$$\Rightarrow \frac{1}{3}v^{-\frac{2}{3}}\frac{dv}{dx} + \frac{1}{3}v^{\frac{1}{3}} = \frac{1}{3}e^{-x}v^{-\frac{2}{3}}$$

$$\frac{dv}{dx} + v = e^{-x}$$

$$\Rightarrow P(x) = 1, Q(x) = e^{-x}$$

$$\Rightarrow f(x) = e^{\int P(x)dx} = e^x$$

$$e^x \frac{dv}{dx} + e^x v = e^x e^{-x}$$

$$\frac{d}{dx}[e^x v] = 1$$

$$e^x v = \int 1 dx$$

$$= x + C$$

$$\Rightarrow v(x) = e^{-x}(x + C)$$

$$y(x) = (v(x))^{\frac{1}{3}}$$

$$= e^{-\frac{x}{3}}(x + C)^{\frac{1}{3}}$$

P31. solution: $(2x+3y)dx + (3x+2y)dy = 0$

$$M(x,y) = 2x+3y, N(x,y) = 3x+2y$$

$$\frac{\partial M}{\partial y} = 3, \frac{\partial N}{\partial x} = 3$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \text{exact!}$$

$$\frac{\partial F}{\partial x} = M(x,y) \Rightarrow$$

$$F(x,y) = \int M(x,y)dx = \int (2x+3y)dx$$

$$= x^2 + 3xy + g(y)$$

$$\frac{\partial F}{\partial y} = N(x,y) \Rightarrow$$

$$\frac{\partial F}{\partial y} = 3x + g'(y) = 3x + 2y$$

$$\Rightarrow g'(y) = 2y$$

$$\Rightarrow g(y) = \int 2y dy = y^2 + C_1$$

Thus $F(x,y) = x^2 + 3xy + y^2 + C$,
and a general solution is implicitly
defined by $x^2 + 3xy + y^2 = C$.

P36. solution: $(1+y e^{xy})dx + (2y + x e^{xy})dy = 0$

$$M(x,y) = 1 + y e^{xy}, N(x,y) = 2y + x e^{xy}$$

$$\frac{\partial M}{\partial y} = x e^{xy}, \frac{\partial N}{\partial x} = x e^{xy}$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \text{exact!}$$

$$\frac{\partial F}{\partial x} = M(x,y) \Rightarrow$$

$$F(x,y) = \int M(x,y)dx = \int (1 + y e^{xy})dx$$

$$= x + e^{xy} + g(y)$$

$$\frac{\partial F}{\partial y} = N(x,y) \Rightarrow$$

$$\frac{\partial F}{\partial y} = x e^{xy} + g'(y) = 2y + x e^{xy}$$

$$\Rightarrow g'(y) = 2y$$

$$\Rightarrow g(y) = \int 2y dy = y^2 + C_1$$

Thus, $F(x,y) = x + e^{xy} + y^2 + C$,
and a general solution is implicitly
given by $x + e^{xy} + y^2 = C$.